

gradient (velocity same direction as displacement).

- 5** **C** While it is true in this example that the weight of the book is equal to the force upwards acting on the book from the table, this is not the pair of forces in Newton's third law. The weight of the book is the gravitational pull of the Earth on the book; the 'reaction' force is the gravitational pull of the book on the Earth. The key points to remember for Newton's third law:

The two forces are

- 1) of the same magnitude (in this case, W)
- 2) in opposite directions (in this case, upwards)
- 3) acts on different objects (in this case, on the Earth)
- 4) of the same type (in this case, gravitational)

- 6** **C** As the ball bounces elastically, no kinetic energy is lost as heat and sound